SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 30**

Annexure A
Resolution Algorithm Implementation

Code of Conduct:

I S. Chandra Shekhar / 192372163 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Chandra Shekhar

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic Representation

Let R = It rains, W = The ground is wet.

Premise 1: $R \rightarrow W$

Premise 2: R

Goal: W

2. Conversion to CNF

$R \rightarrow W \equiv \neg R \lor W$

R (already a unit clause)

3. Negation of Goal

$\neg W$

4. Clause Set

C1: $\neg R \lor W$

C2: R

C3: $\neg W$ (negated goal)

5. Resolution Algorithm – Step by Step

Step 1: Resolve C1 ($\neg R \lor W$) with C3 ($\neg W$) on complementary literal W → derive C4: $\neg R$

Step 2: Resolve C4 ($\neg R$) with C2 (R) on complementary literal R → derive the Empty Clause (□)

6. Conclusion

Since the resolution process derives the Empty Clause (□), the negated goal ¬W leads to a contradiction. Hence, the goal W ("the ground is wet") is PROVED to logically follow from the Knowledge Base.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks using the Resolution Algorithm.

1. Propositional Logic Representation

Let S = Rahul submits the assignment, M = Rahul receives internal marks.

Premise 1: S → M

Premise 2: S

Goal: M

2. Conversion to CNF

S → M ≡ ¬S ∨ M

S (unit clause)

3. Negation of Goal

¬M

4. Clause Set

C1: ¬S ∨ M

C2: S

C3: ¬M (negated goal)

5. Resolution Algorithm – Step by Step

Step 1: Resolve C1 (¬S ∨ M) with C3 (¬M) on M → derive C4: ¬S
Step 2: Resolve C4 (¬S) with C2 (S) on S → derive the Empty Clause (□)

6. Conclusion

The derivation of the Empty Clause (□) shows that ¬M is a contradiction. Hence the goal M ("Rahul receives internal marks") is PROVED.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books using the Resolution Algorithm.

1. Propositional Logic Representation

Let L = Priya is a library member, B = Priya can borrow books.

Premise 1: L → B

Premise 2: L

Goal: B

2. Conversion to CNF

L → B ≡ ¬L ∨ B

L (unit clause)

3. Negation of Goal

¬B

4. Clause Set

C1: $\neg L \lor B$

C2: L

C3: $\neg B$ (negated goal)

5. Resolution Algorithm – Step by Step

Step 1: Resolve C1 ($\neg L \lor B$) with C3 ($\neg B$) on B → derive C4: $\neg L$

Step 2: Resolve C4 ($\neg L$) with C2 (L) on L → derive the Empty Clause (□)

Step 3: No new clauses can be generated after this; the algorithm terminates.

6. Conclusion

Since the Empty Clause (□) is derived, the negation of the goal is contradictory. Hence B ("Priya can borrow books") is PROVED to be true.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement using the Resolution Algorithm.

1. Propositional Logic Representation

Let A = Arun clears the aptitude test, E = Arun is eligible for placement.

Premise 1: $A \rightarrow E$

Premise 2: A

Goal: E

2. Conversion to CNF

$A \rightarrow E \equiv \neg A \lor E$

A (unit clause)

3. Negation of Goal

$\neg E$

4. Clause Set

C1: $\neg A \lor E$

C2: A

C3: $\neg E$ (negated goal)

5. Resolution Algorithm – Step by Step

Step 1: Resolve C1 ($\neg A \lor E$) with C3 ($\neg E$) on E → derive C4: $\neg A$

Step 2: Resolve C4 ($\neg A$) with C2 (A) on A → derive the Empty Clause (□)

6. Conclusion

Resolution Tree: $\{\neg A \lor E\} + \{\neg E\} \rightarrow \{\neg A\}$; $\{\neg A\} + \{A\} \rightarrow$ □. Since the Empty Clause is reached, E ("Arun is eligible for placement") is PROVED.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

1. Propositional Logic Representation

Let P = User enters correct password, Au = User is authenticated, G = User is granted access.

Premise 1: $P \rightarrow Au$

Premise 2: $Au \rightarrow G$

Premise 3: P

Goal: G

2. Conversion to CNF

$P \rightarrow Au \equiv \neg P \lor Au$

$Au \rightarrow G \equiv \neg Au \lor G$

P (unit clause)

3. Negation of Goal

$\neg G$

4. Clause Set

C1: $\neg P \lor Au$

C2: $\neg Au \lor G$

C3: P

C4: $\neg G$ (negated goal)

5. Resolution Algorithm – Step by Step

Step 1: Resolve C1 ($\neg P \lor Au$) with C3 (P) on P → derive C5: Au

Step 2: Resolve C5 (Au) with C2 ($\neg Au \lor G$) on Au → derive C6: G

Step 3: Resolve C6 (G) with C4 ($\neg G$) on G → derive the Empty Clause (□)

6. Conclusion

The step-by-step resolution shows that P (correct password) implies Au (authenticated), which implies G (granted access). The derivation of the Empty Clause (□) confirms the negated goal is contradictory. Hence G ("the user is granted access") is PROVED.